

Fig. 2: An oscilloscope represents a graph of signals amplitude versus time

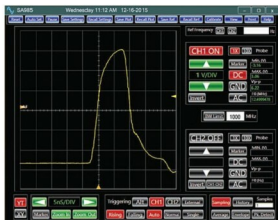


Fig. 3: Display of a 10ns pulse with a 1GHz bandwidth oscilloscope

acquired. This way the user will not be limited by traditional triggers.”

Jitter analysis of AOCs

Evaluation of active optical cables (AOCs) not only requires bit error rate (BER) measurements and eye-pattern analyses but also requires jitter analysis due to analysis at electrical signal conversion. “By using MP2100B along with jitter analysis software MX210001A option, jitter of high-speed, multichannel AOCs can be measured at the same time as performing eye-pattern and eye-mask tests.

Moreover, adoption of high-speed trigger methods supports high-speed data dependent jitter (DDJ) measurements, cutting measurement times by 80 per cent compared to legacy models, thereby cutting overall measurement times,” explains Madhukar Tripathi, senior

manager - marketing and channel sales, Anritsu India Pvt Ltd.

Testing for newer standards

Better signal integrity performance, noise performance, lower intrinsic data and more stable time base with higher accuracy are some of the improvements seen in recent launches of oscilloscopes.

People who need high signal integrity and are working on memory technology like double data rate type three (DDR3) can use scopes like the S-series.

The V-series features 160-bit serial hardware trigger with data rates of 12.5 gigasamples per second and a mixed-signal oscilloscope of up to 20 gigasamples per second, which is capable of any-

thing from 8GHz to 33GHz.

“This is best for DDR4, LPDDR4 and USB 3.0. You need high-serial trigger capability to deal with technology like PCI Express 3.0, which is symbol-coded with 130 symbols. The same goes for other symbol-coded technologies like USB 3.1,” adds Srivastava.

Oscilloscopes with a high time accuracy of 12 parts per million have very low intrinsic data. This lets engineers using measurements like eye diagrams to have more margins available for them. Your device under test has a greater chance of succeeding here while it may fail while testing on other equipment.

The last 18 months have seen support being added or improved for USB 3.1, High-Definition Multimedia Interface (HDMI) 2.0, DisplayPort 1.2 and Ethernet

protocol decoder. Also, compliance applications for 10GB and 100GB Ethernet used in backplanes were made available.

Equipment like BERTWave MP2100B can perform simultaneous BER measurements and eye-pattern analyses, and thus both equipment investment costs and test times are halved.

“Additionally, the tracking function makes it easy to set both BERT and eye/pulse scope functions. Using the built-in all BER result function supports simultaneous BER measurement of four channels, cutting measurement time to less than one-third that of legacy models,” explains Tripathi.

Better add-ons

Keysight launched a new probe using which engineers are able to perform power-integrity measurements. Srivastava explains that, “Earlier we did not have a dedicated probe, so you had to use Ethernet cables to connect directly. The new probe has a high bandwidth of 2GHz and a low attenuation ratio of 1:1, which has a reduced effect on the signal. It only adds ten per cent to the baseline noise of the oscilloscope that it is attached to.”

National Instruments has also brought out new probes that can handle up to ± 300 volts without the scope needing an inbuilt attenuator.

In the last 18 months, Infinium firmware has also been completely revamped. It now delivers a new look and allows users to customise displays, thus letting them create multiple displays, eight waveforms, tabs and a custom layout. It now also includes the ability to launch an eye diagram with one click on a waveform (you previously had to configure it).

Form factors

Form factors for devices like the S-series from Keysight have mostly remained the same despite the sub-